

Companion XIV: Primordial Black Holes as CPT-Sector Objects in the Relaxation-Driven Cyclic Cosmology (RDCC)

Michael Lehmann¹

¹Independent Researcher, Relaxation-Driven Cyclic Cosmology (RDCC) Framework,
mi.lehmann@gmx.de

June 7, 2026

Abstract

Primordial black holes (PBHs) provide a uniquely sensitive probe of early-Universe physics, constraining the thermal history, the growth of perturbations, and the collapse thresholds during radiation domination. In the Relaxation-Driven Cyclic Cosmology (RDCC), the bipartite CPT-symmetric global state, the sectoral temperature asymmetry, and the universal relaxation mechanism jointly reshape the conditions for PBH formation. In this Companion we develop a sectoral interpretation of PBHs within RDCC: PBHs arise predominantly in the CPT-conjugate (mirror) sector, where the lower temperature, modified relaxation history, and altered perturbation dynamics naturally enhance PBH formation. The visible sector, by contrast, exhibits a strong suppression of PBHs due to relaxation-induced damping and an elevated effective collapse threshold. We derive the CPT-sector PBH formation mechanism, compute the resulting mass function, and identify the observational signatures associated with this sectoral asymmetry. This Companion establishes PBHs as a natural and robust component of the CPT sector and provides a new observational window into the bipartite structure and relaxation-driven dynamics of RDCC.

1 Introduction

Primordial black holes (PBHs) provide a uniquely sensitive probe of the earliest stages of cosmic evolution. Their formation depends on the thermal history, the growth of primordial perturbations, and the collapse thresholds during radiation domination. In standard cosmology, PBHs require either enhanced primordial fluctuations, non-Gaussian features, or exotic early-time dynamics to form in appreciable abundance.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally different. The bipartite CPT-symmetric global state, the sectoral temperature asymmetry, and the relaxation-induced modification of perturbation dynamics jointly reshape the conditions for PBH formation. The relaxation mechanism suppresses the growth of super-horizon perturbations in the visible sector, raising the effective collapse threshold and strongly inhibiting PBH formation. At the same time, the CPT-conjugate sector evolves with a lower temperature, reduced relaxation-induced damping, and a larger horizon mass at a given scale factor.

These structural differences lead to a robust sectoral asymmetry: PBHs form naturally and efficiently in the CPT-conjugate sector, while their formation in the visible sector is exponentially suppressed. This interpretation arises directly from the bipartite global state and the universal relaxation mechanism introduced in Companion X, and it is consistent with the sectoral thermodynamics developed in Companion XIII.

The goal of this Companion is to develop the PBH sector of RDCC in a unified and model-independent manner. Section 2 introduces PBHs as CPT-sector objects and explains the suppression mechanism in the visible sector. Section 3 derives the PBH formation mechanism in the CPT sector. Section 4 develops the modified collapse threshold and the resulting PBH mass function. Section 5 analyzes the thermodynamic and dynamical properties of CPT-sector PBHs. Section 6 presents the observational signatures, and Section 7 provides forecasts for future surveys.

Together, these results establish PBHs as a natural and robust component of the CPT sector and provide a new observational window into the bipartite structure and relaxation-driven dynamics of RDCC.

2 PBHs as CPT-Sector Objects

The Relaxation-Driven Cyclic Cosmology (RDCC) is built on a bipartite, CPT-symmetric global state in which two thermodynamically distinct sectors evolve on the same spacetime geometry. A visible-sector observer corresponds to a projection onto the sector with a forward thermodynamic arrow, while the CPT-conjugate sector evolves with decreasing entropy. The two sectors interact only through gravity and the relaxation dynamics.

A key structural result of Companion X is the persistent temperature asymmetry between the sectors,

$$\frac{T_B}{T_A} \simeq 0.5, \quad (1)$$

which remains stable throughout the early Universe. This temperature asymmetry has direct implications for PBH formation, as both the collapse threshold and the horizon mass depend sensitively on the thermal history.

PBH formation requires large density fluctuations to exceed a critical collapse threshold at horizon re-entry. In the visible sector, the relaxation-induced damping of super-horizon perturbations increases the effective collapse threshold, suppressing PBH formation. In the CPT-conjugate sector, the lower temperature and reduced relaxation-induced damping lower the collapse threshold and increase the horizon mass, enhancing PBH formation.

These structural differences lead to three robust conclusions:

- (a) **PBHs are strongly suppressed in the visible sector.** Relaxation-induced damping raises the effective collapse threshold, preventing overdensities from reaching the required amplitude for collapse.
- (b) **PBHs form naturally in the CPT-conjugate sector.** The colder temperature and reduced damping lower the collapse threshold and increase the horizon mass, making PBH formation efficient.
- (c) **PBHs are inherently CPT-sector objects in RDCC.** Their formation mechanism, abundance, and mass spectrum follow directly from the thermodynamic and relaxation properties of the CPT sector.

This interpretation resolves several conceptual tensions. It explains the absence of PBHs in the visible sector without fine-tuning, provides a natural origin for PBHs in the RDCC framework, and connects PBH physics directly to the bipartite structure of the global state.

In the next section, we derive the CPT-sector PBH formation mechanism and quantify the degree of suppression in the visible sector.

3 CPT-Sector PBH Formation Mechanism

Primordial black holes (PBHs) form when large density fluctuations re-enter the horizon during radiation domination and exceed a critical collapse threshold. In standard cosmology, this threshold is typically of order

$$\delta_{\text{std}} \sim 0.4 - 0.5, \quad (2)$$

and PBH formation requires either enhanced primordial fluctuations or non-Gaussian features.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the early-Universe dynamics are modified by the universal relaxation mechanism introduced in Companion X. The relaxation rate $\Gamma(a) = \alpha(a)H(a)$ alters the growth of super-horizon perturbations and increases the effective collapse threshold in the visible sector. As a result, PBH formation in the visible sector is strongly suppressed.

In the CPT-conjugate sector, the situation is fundamentally different. The lower temperature, reduced relaxation-induced damping, and altered perturbation dynamics naturally enhance PBH formation. Three structural ingredients govern the PBH formation mechanism in the CPT sector:

- (a) **Reduced relaxation-induced damping.** The relaxation coefficient satisfies

$$\alpha_B(a) < \alpha_A(a), \quad (3)$$

due to the lower temperature of sector B. This reduces the damping of super-horizon perturbations and lowers the effective collapse threshold.

- (b) **Larger horizon mass.** The colder temperature implies a slower expansion rate and therefore a larger horizon mass at a given scale factor:

$$M_H^{(B)}(a) \simeq 4 M_H^{(A)}(a). \quad (4)$$

This shifts the PBH mass spectrum in sector B toward higher masses.

- (c) **Lower collapse threshold.** The reduced pressure in the colder sector lowers the collapse threshold:

$$\delta_c^{(B)} < \delta_c^{(A)}. \quad (5)$$

This exponentially enhances PBH formation in sector B.

Combining these effects yields a natural and robust PBH formation mechanism in the CPT sector. PBHs arise as a direct consequence of the bipartite global state and the relaxation dynamics, without requiring enhanced primordial fluctuations or exotic early-Universe physics.

In contrast, the visible sector experiences a strong suppression of PBH formation due to the elevated collapse threshold and reduced horizon mass. This sectoral asymmetry is a distinctive and testable prediction of RDCC and forms the basis for the modified collapse threshold developed in Section 4.

4 Modified Collapse Threshold in RDCC

Primordial black holes form when large density fluctuations re-enter the horizon during radiation domination and exceed a critical collapse threshold. In standard cosmology, this threshold is of order

$$\delta_{\text{std}} \sim 0.4 - 0.5, \quad (6)$$

and PBH formation requires either enhanced primordial fluctuations or non-Gaussian features.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the early-Universe dynamics are modified by the universal relaxation mechanism introduced in Companion X. The relaxation rate $\Gamma(a) = \alpha(a)H(a)$ alters the growth of super-horizon perturbations and increases the effective collapse threshold in the visible sector. This modification arises from the additional damping term in the perturbation equation of motion.

4.1 Relaxation-Induced Damping

The evolution of super-horizon perturbations in RDCC follows the modified equation

$$\ddot{\delta} + (2H + \Gamma)\dot{\delta} - 2H^2\delta = 0, \quad (7)$$

where the additional damping term $\Gamma(a)$ arises from the inter-sector relaxation dynamics. Using the relation

$$\Gamma(a) = \alpha(a)H(a), \quad (8)$$

the equation becomes

$$\ddot{\delta} + (2 + \alpha)H\dot{\delta} - 2H^2\delta = 0. \quad (9)$$

In the relaxation-dominated regime, $\Gamma \sim \alpha H$, the growth of δ is suppressed by a factor that can be parametrized as an effective enhancement of the collapse threshold. Solving the equation yields the scaling

$$\delta(a) \propto a^{-\alpha/2}, \quad (10)$$

implying that the initial amplitude must be larger by a factor $a^{\alpha/2}$ to reach the standard collapse threshold at horizon re-entry.

4.2 Relaxation-Corrected Collapse Threshold

This motivates the parametrization

$$\delta_c^{(A/B)}(a) = \delta_{\text{std}} \left[1 + \beta \alpha_{A/B}(a) \right], \quad (11)$$

where $\beta = \mathcal{O}(1)$ captures the dependence on the perturbation profile. Because the CPT-conjugate sector is colder, the relaxation coefficient satisfies

$$\alpha_B(a) < \alpha_A(a), \quad (12)$$

and therefore

$$\delta_c^{(B)} < \delta_c^{(A)}. \quad (13)$$

4.3 Exponential Sensitivity of PBH Formation

PBH formation is exponentially sensitive to the collapse threshold. Writing the PBH mass fraction as

$$\beta_{\text{PBH}}^{(A/B)}(M) \propto \exp \left[-\frac{\delta_c^{2(A/B)}}{2\sigma^2(M)} \right], \quad (14)$$

makes clear that even a small relative change

$$\frac{\Delta\delta_c}{\delta_c} \sim \beta \alpha_{A/B}(a), \quad (15)$$

induces an exponentially large difference in β_{PBH} .

This is the origin of the strong sectoral PBH asymmetry in RDCC:

$$\beta_{\text{PBH}}^{(B)} \gg \beta_{\text{PBH}}^{(A)}. \quad (16)$$

The visible sector is exponentially suppressed, while the CPT sector naturally forms PBHs. This sectoral asymmetry is a direct and testable prediction of the RDCC framework and forms the basis for the PBH mass spectrum developed in Section 5.

5 Thermodynamics and Dynamics of CPT-Sector PBHs

PBHs formed in the CPT-conjugate sector inherit the thermodynamic and dynamical properties of that sector. Because the CPT sector evolves with a lower temperature, reduced relaxation coefficient, and modified perturbation dynamics, the resulting PBHs differ systematically from those that would form in the visible sector. Three features are particularly important.

5.1 Temperature Dependence

The colder temperature of sector B implies:

- a slower expansion rate,
- a larger horizon mass at a given scale factor,
- reduced pressure support,
- a lower collapse threshold.

These effects combine to enhance PBH formation relative to the visible sector. The larger horizon mass directly shifts the PBH mass spectrum toward higher masses, while the reduced pressure support lowers the threshold for collapse.

5.2 Relaxation History

The relaxation coefficient satisfies

$$\alpha_B(a) < \alpha_A(a), \quad (17)$$

implying reduced damping of super-horizon perturbations in sector B. This allows overdensities to grow more efficiently prior to horizon re-entry. The relaxation mechanism therefore plays a dual role:

- it suppresses PBH formation in the visible sector by increasing the effective collapse threshold,
- it enhances PBH formation in the CPT sector by reducing damping and lowering the threshold.

This sectoral asymmetry is a direct consequence of the bipartite global state and the universal relaxation dynamics.

5.3 Mass Spectrum and Abundance

The PBH mass spectrum in sector B is shifted toward higher masses due to the larger horizon mass:

$$M_{\text{PBH}}^{(B)} \simeq 4 M_{\text{PBH}}^{(A)}. \quad (18)$$

The abundance is exponentially enhanced:

$$\beta_{\text{PBH}}^{(B)} \propto \exp\left[-\frac{\delta_c^{2(B)}}{2\sigma^2}\right], \quad \delta_c^{(B)} < \delta_c^{(A)}. \quad (19)$$

Together, these features establish PBHs as natural CPT-sector objects in RDCC, arising from the interplay between relaxation dynamics, sectoral thermodynamics, and the bipartite global state.

6 Observational Signatures

The sectoral PBH asymmetry predicted by the Relaxation-Driven Cyclic Cosmology (RDCC) leads to a number of distinctive observational signatures. These signatures arise from the suppression of PBHs in the visible sector, the enhanced PBH abundance in the CPT-conjugate sector, and the shift of the PBH mass spectrum toward higher masses in sector B. Together, they provide a powerful observational handle on the relaxation-driven dynamics that define the RDCC framework.

6.1 Suppressed PBH Abundance in the Visible Sector

The relaxation-induced enhancement of the collapse threshold in sector A,

$$\delta_c^{(A)} = \delta_{\text{std}} [1 + \beta \alpha_A(a)], \quad (20)$$

leads to an exponential suppression of PBH formation. This prediction is consistent with microlensing, CMB, and SGWB constraints, all of which show no evidence for a significant PBH population in the visible sector.

6.2 Enhanced PBH Abundance in the CPT Sector

Because $\alpha_B(a) < \alpha_A(a)$, the collapse threshold in sector B is lower,

$$\delta_c^{(B)} < \delta_c^{(A)}, \quad (21)$$

leading to an exponentially larger PBH abundance. These PBHs are not directly observable, but their existence is a robust prediction of the bipartite global state and the relaxation dynamics.

6.3 Sectoral Mass Asymmetry

The colder temperature of sector B implies a larger horizon mass at collapse,

$$M_{\text{PBH}}^{(B)} \simeq 4 M_{\text{PBH}}^{(A)}, \quad (22)$$

leading to heavier PBHs in the CPT sector. This mass asymmetry is a direct consequence of the sectoral temperature ratio derived in Companion X.

6.4 Connection to Tensor Modulation

The sectoral PBH asymmetry is part of a broader pattern of inter-sectoral imprints in RDCC. Companion II shows that the same bipartite structure and relaxation dynamics induce a modulation of the tensor sector, affecting the primordial gravitational-wave background. Both effects—the CPT-sector PBH population and the tensor modulation—arise from the same underlying inter-sectoral coupling and the temperature asymmetry between the sectors. PBHs therefore provide a complementary probe of the physics that shapes the tensor sector in RDCC.

6.5 Gravitational-Wave Signatures

PBHs contribute to the stochastic gravitational-wave background (SGWB) through mergers and collapse. The suppression in sector A implies:

- a reduced merger rate,
- no PBH-induced SGWB in the visible sector,
- possible indirect signatures from sector B.

These signatures complement the relaxation-induced tensor suppression developed in Companion XV.

6.6 Microlensing Constraints

The strong suppression of PBHs in sector A implies that microlensing surveys should observe no significant PBH population—consistent with current data from EROS, OGLE, Subaru/HSC, and Roman forecasts.

6.7 Modified PBH Mass Spectrum

The relaxation-corrected mass function,

$$\beta_{\text{PBH}}^{(A)}(M) \propto \exp\left[-\frac{\delta_c^2{}^{(A)}}{2\sigma^2(M)}\right], \quad (23)$$

predicts a suppressed and shifted mass spectrum in sector A. The visible sector therefore exhibits neither a significant PBH abundance nor a broad mass distribution.

6.8 Consistency with RDCC Temperature Asymmetry

The sectoral PBH asymmetry provides an independent test of the temperature ratio

$$\frac{T_B}{T_A} \simeq 0.5, \quad (24)$$

derived in Companion X. The PBH sector therefore offers a complementary observational probe of the same relaxation-driven thermodynamics that shape the neutrino and dark matter sectors.

7 Forecasts

The sectoral PBH asymmetry predicted by the Relaxation-Driven Cyclic Cosmology (RDCC) provides multiple observational avenues for testing the framework. Future surveys will be able to probe the PBH sector through gravitational waves, microlensing, and the matter power spectrum. These forecasts highlight the complementarity between PBH physics, tensor observables, and sectoral thermodynamics.

7.1 Gravitational-Wave Forecasts

Space-based interferometers such as LISA, DECIGO, and BBO will be sensitive to the stochastic gravitational-wave background (SGWB) generated by PBH mergers in the CPT sector. While these PBHs are not directly observable, their merger history may leave detectable imprints in the SGWB. The predicted signatures include:

- a sectoral enhancement of the high-frequency SGWB due to heavier PBHs in sector B,
- a merger-driven contribution consistent with the relaxation-induced tensor suppression developed in Companion XV,
- a characteristic frequency dependence linked to the horizon mass in the CPT sector.

These signatures provide an indirect but powerful probe of the CPT-sector PBH population.

7.2 Microlensing Forecasts

Microlensing surveys will continue to tighten constraints on PBHs in the visible sector. The RDCC prediction of strong suppression in sector A is consistent with current data and will be further tested by future surveys such as Roman and LSST. The expected signatures are:

- no detectable PBH population in the visible sector,
- increasingly stringent upper bounds on the PBH mass fraction in sector A,
- consistency with the relaxation-corrected collapse threshold derived in Section 4.

A null detection in microlensing is therefore a natural outcome of the RDCC framework.

7.3 Large-Scale-Structure Forecasts

The absence of PBH-induced anomalies in the matter power spectrum provides an additional test of the RDCC PBH sector. Future surveys such as Euclid and CMB-S4 will be able to probe:

- small-scale suppression consistent with the relaxation-modified perturbation dynamics,
- the absence of PBH-induced isocurvature modes in the visible sector,
- consistency between the PBH sector and the sectoral temperature ratio $T_B/T_A \simeq 0.5$.

These forecasts provide a cross-check between PBH physics, tensor observables, and the relaxation-driven thermodynamics that define the RDCC framework.

8 Conclusion

In this Companion we have developed the primordial black hole (PBH) sector of the Relaxation-Driven Cyclic Cosmology (RDCC), introducing a sectoral interpretation in which PBHs arise naturally and predominantly in the CPT-conjugate sector. The colder temperature, reduced relaxation-induced damping, and modified perturbation dynamics in sector B enhance PBH formation, while the visible sector experiences a strong suppression due to the elevated collapse threshold and reduced horizon mass.

These effects arise directly from the bipartite CPT-symmetric global state and the universal relaxation mechanism introduced in Companion X. The resulting sectoral PBH asymmetry is a robust and testable prediction of RDCC, linking PBH physics to the same relaxation-driven thermodynamics that shape the neutrino and dark matter sectors (Companion XIII) and the tensor sector (Companion XV).

The PBH mass spectrum in the CPT sector is shifted toward higher masses, and the PBH abundance is exponentially enhanced relative to the visible sector. These features provide a distinctive observational target, with signatures in the stochastic gravitational-wave background, microlensing constraints, and the matter power spectrum. Future surveys such as LISA, DECIGO, BBO, Roman, LSST, Euclid, and CMB-S4 will be able to probe these predictions.

With the development of the PBH sector, RDCC now offers a coherent and internally consistent description of early-Universe thermodynamics, perturbation dynamics, and sectoral structure. PBHs emerge as a natural and robust component of the CPT sector and a key probe of relaxation-driven cosmology.